

CUSTOMER SALES ANALYSIS REPORT

1. Project Overview

Customer Sales Analysis

This project analyzes customer purchasing patterns using Python and the Pandas library. The objective is to identify valuable customers, understand regional sales performance, analyze product sales, and generate business insights through data analysis and visualization.

Objectives

- Analyze customer purchasing behavior
- Identify top customers
- Analyze regional sales performance
- Find best-selling products
- Create pivot tables for business insights
- Build professional visualizations
- Generate recommendations for improving business performance

2. Dataset Description

Two datasets were used for this project.

Dataset 1: sales_data.csv

Column	Description
Date	Purchase date
Product	Product purchased
Quantity	Quantity sold

Column	Description
Price	Price per product
Customer_ID	Customer identifier
Region	Sales region
Total_Sales	Total sales amount

Number of Records: **100**

Dataset 2: customer_churn.csv

Column	Description
CustomerID	Customer ID
Tenure	Customer relationship duration
MonthlyCharges	Monthly charges
TotalCharges	Total charges
Contract	Contract type
PaymentMethod	Payment method
PaperlessBilling	Billing type
SeniorCitizen	Senior citizen indicator
Churn	Customer churn status

Number of Records: **500**

3. Data Cleaning Steps

The following preprocessing steps were performed:

- Loaded datasets using Pandas.
- Verified dataset structure using `info()`.
- Checked the first five rows using `head()`.
- Generated descriptive statistics using `describe()`.
- Checked missing values using `isnull().sum()`.
- Converted the **Date** column into datetime format.
- Created new columns:
 - Year
 - Month
- Checked duplicate records.
- Removed duplicate rows (none found).

Result:

- Missing Values: **0**
- Duplicate Records: **0**

4. Customer Analysis

Total Revenue

12,365,048

Total Customers

100

Top 10 Customers

Customer ID	Total Sales
CUST016	373,932
CUST007	363,870
CUST083	350,888
CUST073	349,510
CUST020	333,992
CUST084	324,144
CUST070	318,762
CUST005	318,680
CUST065	312,564
CUST028	304,465

Customer Insights

- Top customer generated 373,932 in sales.
- Revenue is spread across many customers.
- High-value customers should receive loyalty rewards and personalized offers.

5. Sales Analysis

Regional Sales

Region	Total Sales
North	3,983,635
South	3,737,852
East	2,519,639

Region Total Sales

West 2,123,922

Highest Revenue Region:

North

Product Sales

Product Total Sales

Laptop 3,889,210

Tablet 2,884,340

Phone 2,859,394

Headphones 1,384,033

Monitor 1,348,071

Best Selling Product:

Laptop

6. Pivot Table Results

A pivot table was created to summarize sales by **Region** and **Product**.

The pivot table helps identify:

- Highest-performing products in each region.
- Product demand across different locations.
- Regional sales trends.

```
C:\WINDOWS\system32\cmd. x + v
Highest Revenue Region: North
Best Selling Product: Laptop
Month
April      1103468
February   2656050
January    4120524
March      4085006
Name: Total_Sales, dtype: int64
Month
April      44
February   112
January    147
March      175
Name: Quantity, dtype: int64
Product
Headphones 28692.133333
Laptop     27651.500000
Monitor    20709.666667
Phone      27379.000000
Tablet     24177.230769
Name: Price, dtype: float64
Most Expensive Product: Headphones
Product    Headphones  Laptop  Monitor  Phone  Tablet
Region
East        288361    221946    642870    506828    859634
North       107091    1798206    397100    489284    1191954
South       512168    1373120    39924    1471428    341212
West        476413    495938    268177    391854    491540

CUSTOMER SALES ANALYSIS REPORT
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Total Revenue: 12365048
Total Customers: 100
Highest Revenue Region: North
Best Selling Product: Laptop

Top Customer
Customer_ID
CUST016    373932
Name: Total_Sales, dtype: int64
Press any key to continue . . .
```

7. Visualizations

Insert screenshots of the following charts:

Figure 1

Top 10 Customers (Bar Chart)

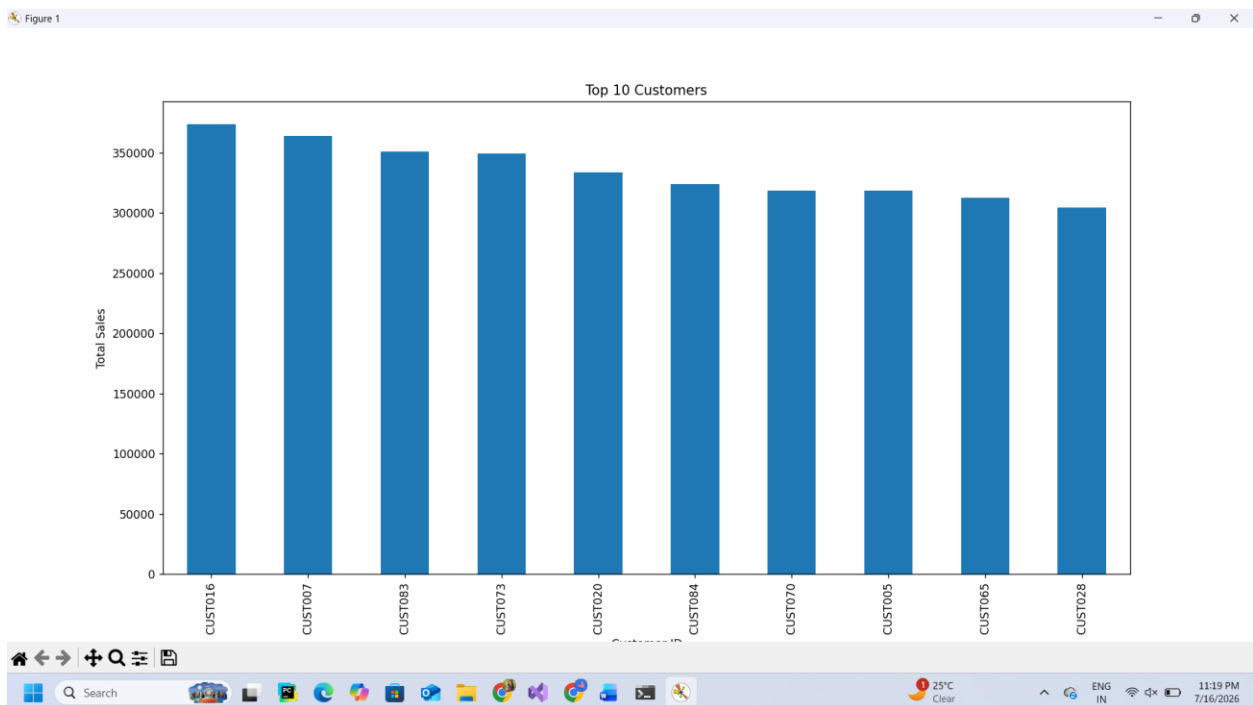


Figure 2

Regional Sales (Bar Chart)

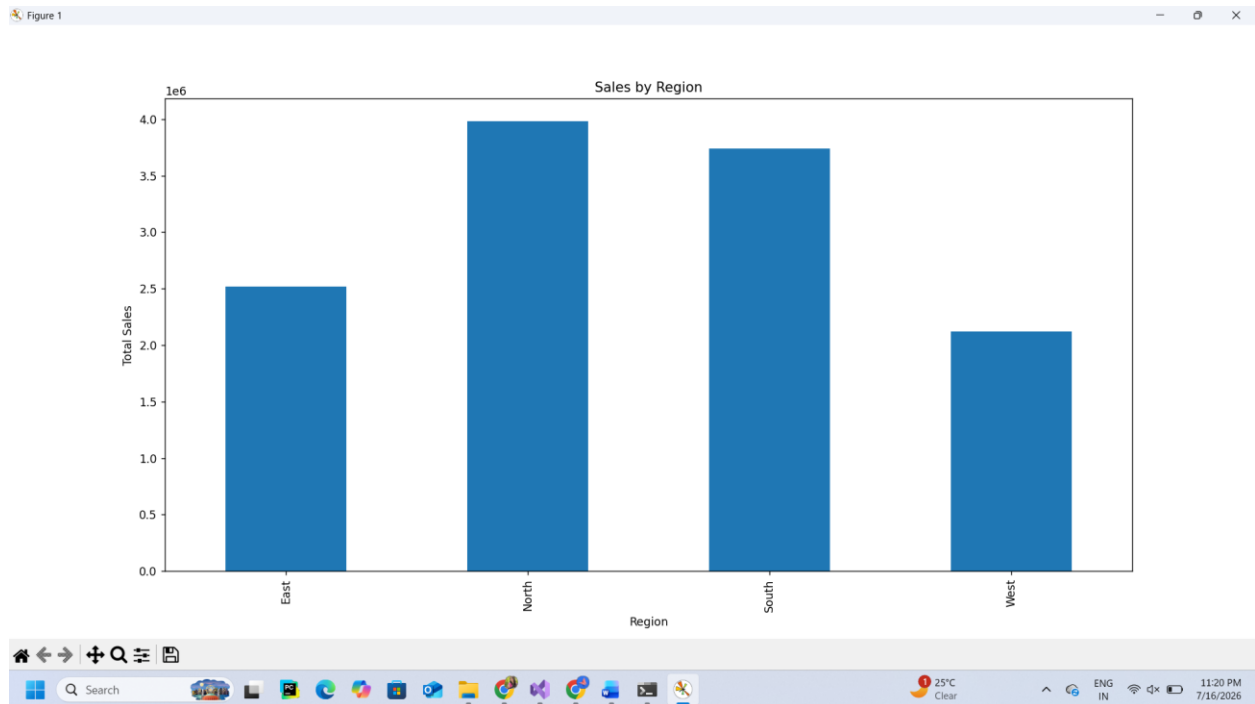


Figure 3

Product Sales Distribution (Pie Chart)

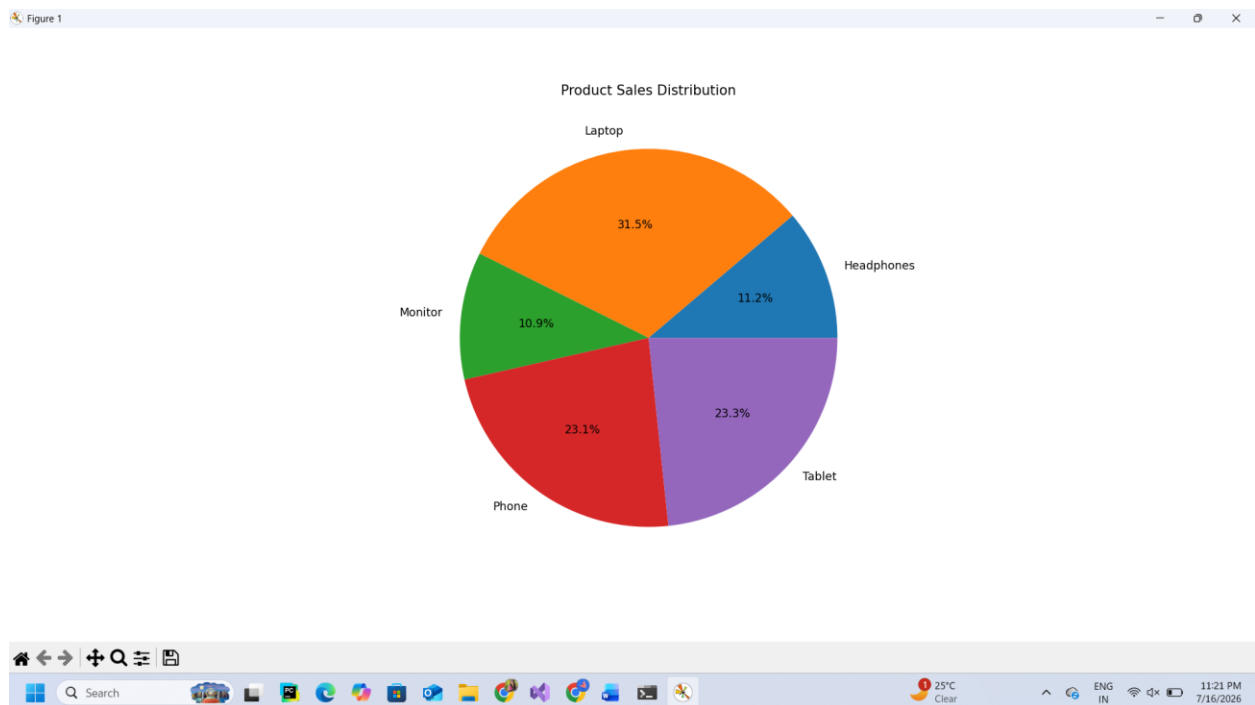


Figure 4

Quantity Sold by Product (Horizontal Bar Chart)

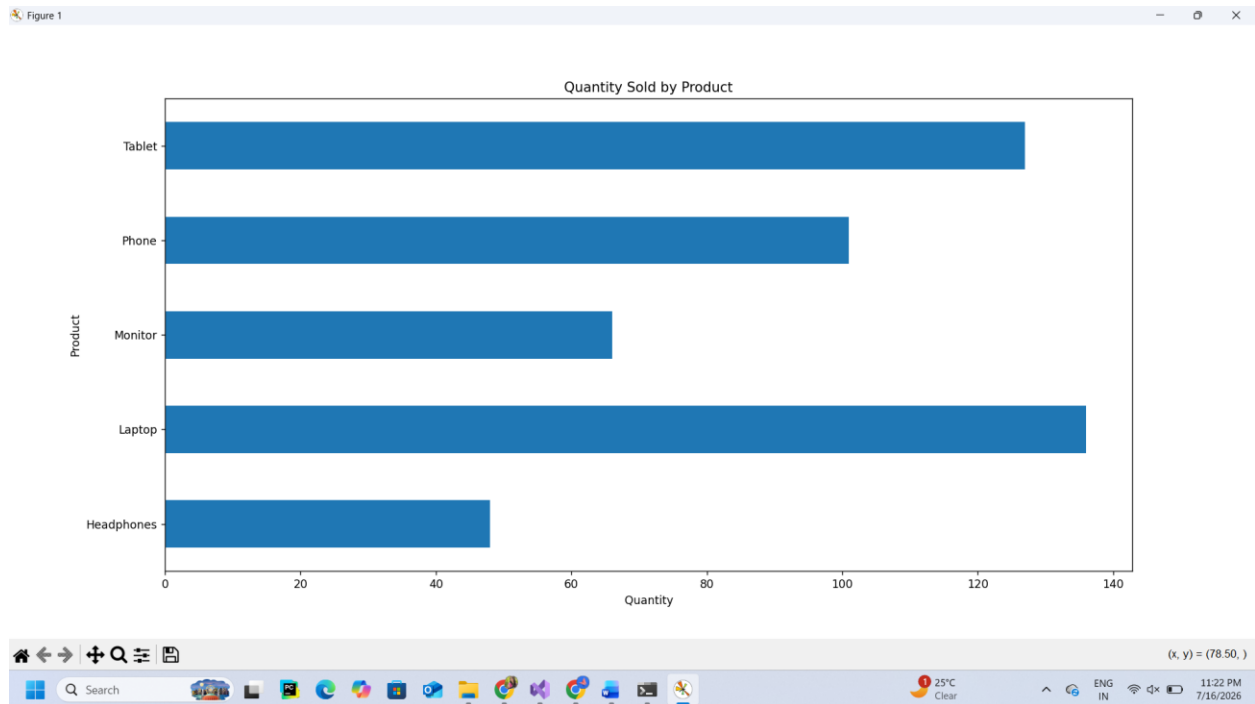
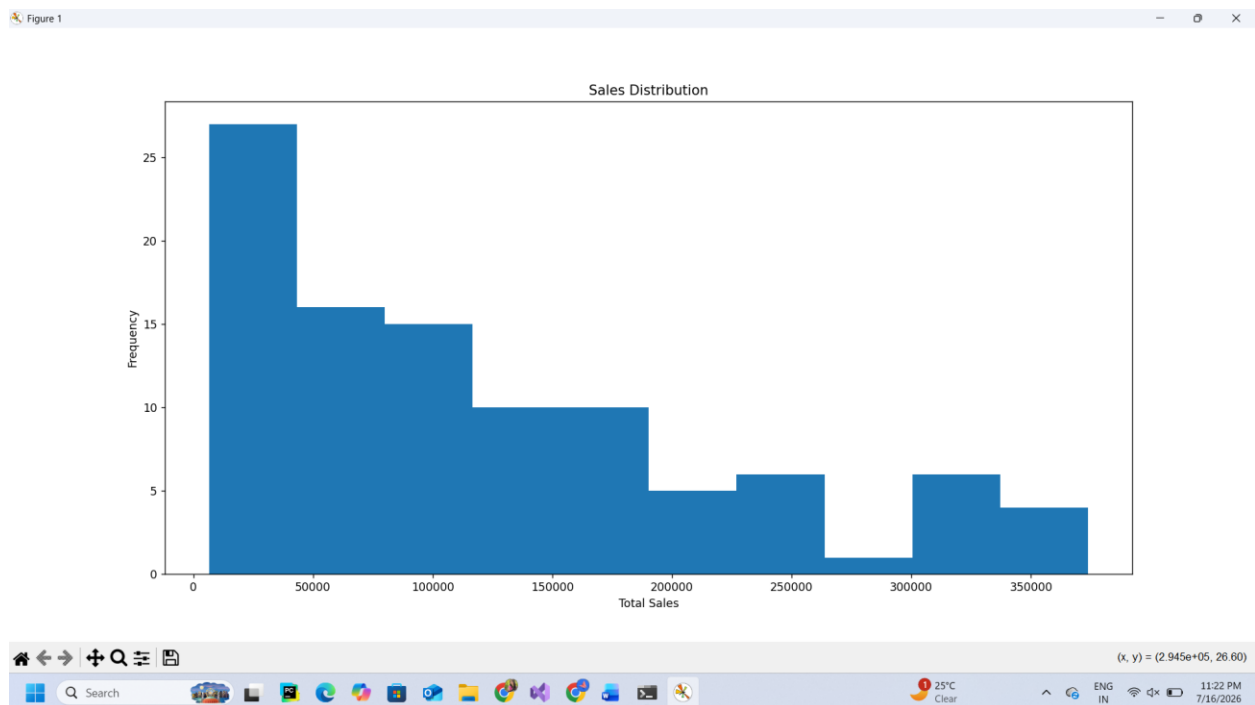


Figure 5

Sales Distribution (Histogram)



8. Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Focus marketing campaigns in the **North** region, which generated the highest revenue.
2. Increase inventory for **Laptops**, as they are the highest-selling product.
3. Introduce loyalty programs for top customers to improve retention.
4. Promote lower-selling products through discounts and bundle offers.
5. Analyze customer purchasing behavior regularly to identify future sales trends.
6. Improve customer engagement using personalized promotions.

9. Executive Summary

The Customer Sales Analysis project successfully analyzed sales transactions using Python and Pandas.

Key findings include:

- Total Revenue: **12,365,048**
- Total Customers: **100**
- Highest Revenue Region: **North**
- Best Selling Product: **Laptop**
- Top Customer: **CUST016**

The project demonstrated data loading, cleaning, aggregation, pivot table creation, customer analysis, sales analysis, and visualization. The generated dashboard and business insights provide valuable information for strategic decision-making and future business growth.